

Module 5: Assignment 1- VPC creation

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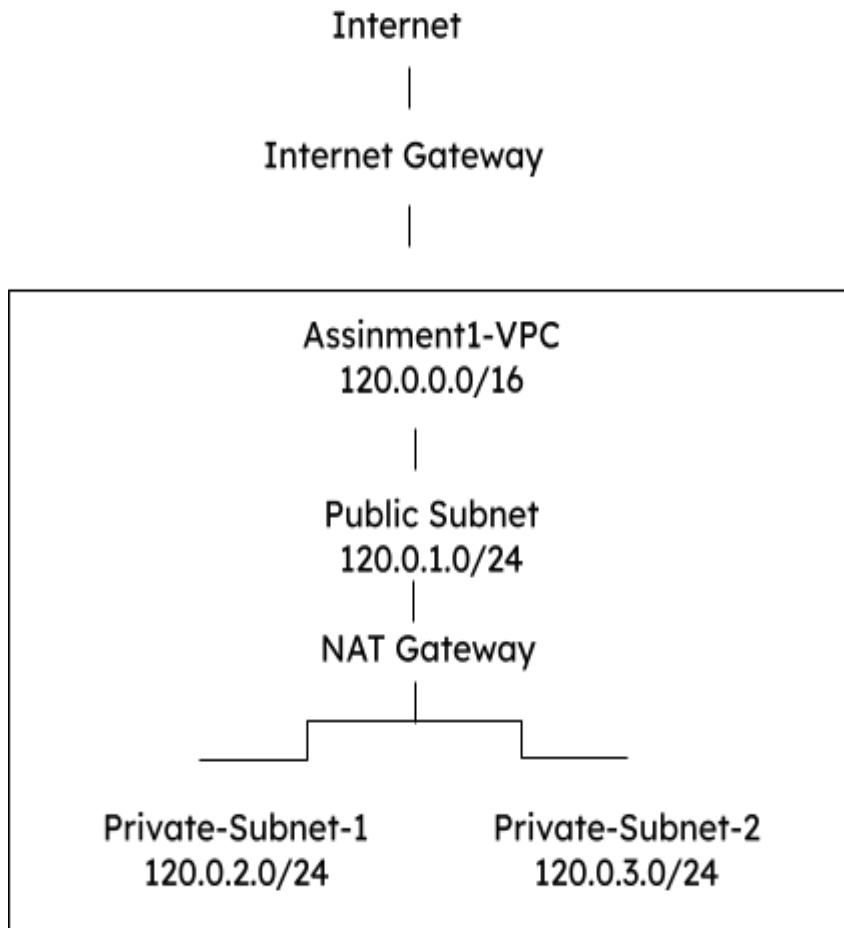
Problem Statement:

“Working for an organization, you are required to provide them a safe and secure environment for the deployment of their resources. They might require different types of connectivity. Implement the following to fulfill the requirements of the company.”

Tasks To Perform:

1. Create a VPC with 120.0.0.0/16 CIDR block.
2. Create 1 public subnet 2 private subnets and make sure you connect a NAT gateway for internet connectivity to a private subnet.

Architecture:



Solution Steps:

1. VPC created

AWS Console → VPC → Your VPCs → Create VPC

Selected:

- **Resources to create:** VPC only
- **Name tag:** Assignment1-VPC
- **IPv4 CIDR:** 120.0.0.0/16
- IPv6: No IPv6 CIDR block
- Tenancy: Default

Clicked **Create VPC**.

VPC > Your VPCs > Create VPC

Create VPC

Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create

Info

Create only the VPC resource or the VPC and other networking resources.

VPC only

VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

Assignment1-VPC

IPv4 CIDR block

Info

IPv4 CIDR manual input

IPAM-allocated IPv4 CIDR block

IPv4 CIDR

120.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block

Info

No IPv6 CIDR block

IPAM-allocated IPv6 CIDR block

Amazon-provided IPv6 CIDR block

IPv6 CIDR owned by me

Tenancy

Info

Default

VPC encryption control (\$)

Info

Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. Additional charges apply

None

Monitor mode

Enforce mode

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q Assignment1-VPC X

Remove tag

Add tag

You can add 49 more tags

Cancel

Preview code

Create VPC

VPC > Your VPCs > vpc-0bc6fabf7f2bc9905

You successfully created vpc-0bc6fabf7f2bc9905 / Assignment1-VPC

X

vpc-0bc6fabf7f2bc9905 / Assignment1-VPC

Actions

Details

Info

VPC ID

vpc-0bc6fabf7f2bc9905

DNS resolution

Enabled

Main network ACL

acl-02bf1922231a971b1

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-0283251ea98d3194b

IPv4 CIDR

120.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-0320e320755e5f562

IPv6 pool

-

Owner ID

430776404757

Successfully created: **Assignment1-VPC** , **CIDR: 120.0.0.0/16**

2. Public-subnet created

VPC → Subnets → Create subnet

Selected your VPC: Assignment1-VPC

For subnet:

- Subnet name: Public-Subnet
- Availability Zone: us-east-1a
- IPv4 subnet CIDR: 120.0.1.0/24

Click **Create subnet**.

The screenshot displays the AWS Management Console interface for a subnet named 'Public-Subnet' with ID 'subnet-06759ad64893e9c2a'. The console shows various configuration details organized into columns:

- Subnet ID:** subnet-06759ad64893e9c2a
- Subnet ARN:** arn:aws:ec2:us-east-1:430776404757:subnet/subnet-06759ad64893e9c2a
- State:** Available
- Block Public Access:** Off
- IPv4 CIDR:** 120.0.1.0/24
- Available IPv4 addresses:** 251
- IPv6 CIDR:** -
- IPv6 CIDR association ID:** -
- Availability Zone:** use1-az1 (us-east-1a)
- Network border group:** us-east-1
- VPC:** vpc-0bc6fabf7f2bc9905 | Assignment1-VPC
- Route table:** -
- Network ACL:** -
- Default subnet:** No
- Auto-assign public IPv4 address:** No
- Auto-assign IPv6 address:** No
- Auto-assign customer-owned IPv4 address:** No
- Customer-owned IPv4 pool:** -
- Outpost ID:** -
- IPv6 CIDR reservations:** -
- Resource name DNS AAAA record:** Disabled
- DNS64:** Disabled
- Hostname type:** IP name
- Owner:** 430776404757
- Resource name DNS A record:** Disabled

Current structure:

Assignment1-VPC

└─ Public-Subnet

└─ 120.0.1.0/24

3. Enable auto assign public IPv4 for Public-subnet:

Actions → Edit subnet settings → Enable auto-assign public IPv4 address → Save

Edit subnet settings
Info

Subnet

Subnet ID

subnet-06759ad64893e9c2a

Name

Public-Subnet

Auto-assign IP settings

Info

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address
Info

☐ Enable auto-assign customer-owned IPv4 address
Info

Option disabled because no customer owned pools found.

VPC
Subnets
subnet-06759ad64893e9c2a

You have successfully changed subnet settings:

- Enable auto-assign public IPv4 address

subnet-06759ad64893e9c2a / Public-Subnet

Details

Subnet ID

subnet-06759ad64893e9c2a

IPv4 CIDR

120.0.1.0/24

Availability Zone

use1-az1 (us-east-1a)

Network ACL

acl-02bf1922231a971b1

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

–

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-1:430776404757:subnet/subnet-06759ad64893e9c2a

Available IPv4 addresses

251

Network border group

us-east-1

Default subnet

No

Customer-owned IPv4 pool

–

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

–

VPC

vpc-0bc6fabf7f2bc9905 | Assignment1-VPC

Auto-assign public IPv4 address

Yes

Outpost ID

–

Hostname type

IP name

Owner

430776404757

Block Public Access

Off

IPv6 CIDR association ID

–

Route table

rtb-0320e320755e5f562

Auto-assign IPv6 address

No

IPv4 CIDR reservations

–

Resource name DNS A record

Disabled

Successfully enabled auto-assign public IPv4 address.

4. A. Private-subnet-1 created

VPC → Subnets → Create subnet

Used:

- Subnet name: Private-Subnet-1
- Availability Zone: us-east-1a
- IPv4 CIDR: 120.0.2.0/24

subnet-03e06ff5a7718f1d4 / Private-Subnet-1			
Details			
Subnet ID subnet-03e06ff5a7718f1d4	Subnet ARN arn:aws:ec2:us-east-1:430776404757:subnet/subnet-03e06ff5a7718f1d4	State Available	Block Public Access Off
IPv4 CIDR 120.0.2.0/24	Available IPv4 addresses 251	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone use1-az1 (us-east-1a)	Network border group us-east-1	VPC vpc-0bc6fabf7f2bc9905 Assignment1-VPC	Route table rtb-0320e320755e5f562
Network ACL acl-02bf1922231a971b1	Default subnet No	Auto-assign public IPv4 address No	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -	IPv4 CIDR reservations -
IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 430776404757	

4. B. Private-subnet-2 created in another AZ

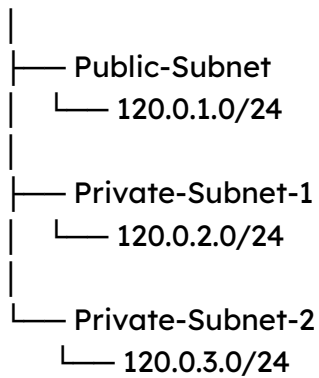
Created another subnet:

- Subnet name: Private-Subnet-2
- Availability Zone: us-east-1b
- CIDR: 120.0.3.0/24

subnet-0b38a8a15ee47400c / Private-Subnet-2			
Details			
Subnet ID subnet-0b38a8a15ee47400c	Subnet ARN arn:aws:ec2:us-east-1:430776404757:subnet/subnet-0b38a8a15ee47400c	State Available	Block Public Access Off
IPv4 CIDR 120.0.3.0/24	Available IPv4 addresses 251	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone use1-az1 (us-east-1a)	Network border group us-east-1	VPC vpc-0bc6fabf7f2bc9905 Assignment1-VPC	Route table rtb-0320e320755e5f562
Network ACL acl-02bf1922231a971b1	Default subnet No	Auto-assign public IPv4 address No	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -	IPv4 CIDR reservations -
IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 430776404757	

Current structure:

Assignment1-VPC



5. Internet Gateway created

VPC → Internet gateways → Create internet gateway

Name: **Assignment1-IGW** → Clicked **Create internet gateway**.

The screenshot shows the AWS Management Console page for the internet gateway **igw-0ab8f36510b5ecbee / Assignment1-IGW**. A green notification banner at the top states: "The following internet gateway was created: igw-0ab8f36510b5ecbee - Assignment1-IGW. You can now attach to a VPC to enable the VPC to communicate with the internet." Below this, the details table shows:

Details	Info
Internet gateway ID	igw-0ab8f36510b5ecbee
State	Detached
VPC ID	-
Owner	430776404757

Selected → **Actions** → **Attach to a VPC**.

selected: Assignment1-VPC → Clicked **Attach internet gateway**.

The screenshot shows the "Attach to VPC (igw-0ab8f36510b5ecbee)" dialog box. It includes a "VPC" section with instructions to attach the gateway to a VPC. Under "Available VPCs", the VPC **vpc-0bc6fab7f2bc9905** is selected. At the bottom, there is an "AWS Command Line Interface command" section and two buttons: "Cancel" and "Attach internet gateway".

The screenshot shows the updated details of the internet gateway **igw-0ab8f36510b5ecbee / Assignment1-IGW**. The details table now shows:

Details	Info
Internet gateway ID	igw-0ab8f36510b5ecbee
State	Attached
VPC ID	vpc-0bc6fab7f2bc9905 Assignment1-VPC
Owner	430776404757

Current structure:

Internet
|
IGW
|
Assignment1-VPC

6. Route table created for the public subnet

VPC → Route tables → Create route table

Enter:

- Name: Public-RT
- VPC: Assignment1-VPC

Next : selected **Public-RT**.

Routes → Edit routes → Add route

Enter:

- Destination: 0.0.0.0/0
- Target: Internet Gateway
- Assignment1-IGW

Clicked **Save changes**



Public-RT

120.0.0.0/16 → local

0.0.0.0/0 → Assignment1-IGW

Routes (2)

Both Edit routes

Filter routes

< 1 > ⚙

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0ab8f36510b5ecbee	Active	No	Create Route
120.0.0.0/16	local	Active	No	Create Route Table

7. Associate Public-RT with the public subnet

VPC → Route tables → selected **Public-RT**
inside **Public-RT**:

Subnet associations → Edit subnet associations

rtb-04df88f01e606c973 / Public-RT

Actions

Details Info

Route table ID
rtb-04df88f01e606c973

VPC
vpc-0bc6fabf7f2bc9905 | Assignment1-VPC

Main
No

Owner ID
430776404757

Explicit subnet associations
-

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Explicit subnet associations (0)

Edit subnet associations

Find subnet association

< 1 > ⚙

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
No subnet associations You do not have any subnet associations.			

Selected: Public-Subnet → Clicked **Save associations**.

VPC > Route tables > rtb-04df88f01e606c973 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/3)

Filter subnet associations

< 1 > ⚙

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public-Subnet	subnet-06759ad64893e9c2a	120.0.1.0/24	-	Main (rtb-0320e320755e5f562)
☐	Private-Subnet-2	subnet-0b38a8a15ee47400c	120.0.3.0/24	-	Main (rtb-0320e320755e5f562)
☐	Private-Subnet-1	subnet-03e06ff5a7718f1d4	120.0.2.0/24	-	Main (rtb-0320e320755e5f562)

Selected subnets

subnet-06759ad64893e9c2a / Public-Subnet

Cancel Save associations

The public subnet successfully has Internet routing.

8. Create the NAT Gateway

VPC → NAT gateways → Create NAT gateway

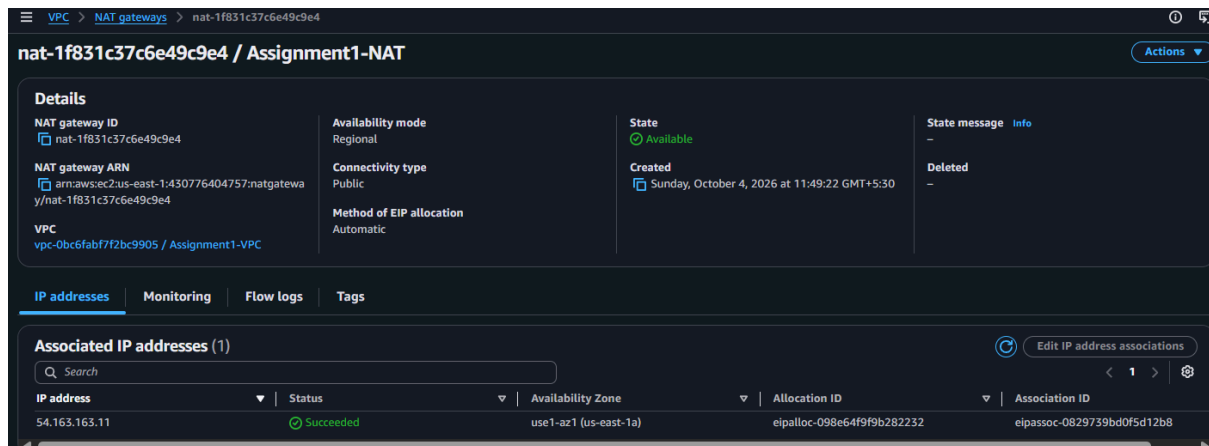
Name: Assignment1-NAT

Subnet: Public-Subnet

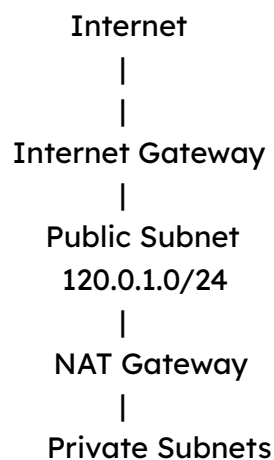
Connectivity type: Public

For **Elastic IP allocation**, click: Allocate Elastic IP

Clicked: Create NAT gateway



Current architecture:



9. Private route table create

VPC → Route tables → Create route table

Config:

Name: Private-RT

VPC: Assignment1-VPC

Route table rtb-0dc7c8a6ce02c09a3 | Private-RT was created successfully.

rtb-0dc7c8a6ce02c09a3 / Private-RT

Details Info

Route table ID
rtb-0dc7c8a6ce02c09a3

VPC
vpc-0bc6fabf7f2bc9905 | Assignment1-VPC

Main
No

Owner ID
430776404757

Explicit subnet associations
-

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (1)

Filter routes

Destination	Target	Status	Propagated	Route Origin
120.0.0.0/16	local	Active	No	Create Route Table

10. NAT route added

Selected **Private-RT**.

Routes → Edit routes → Add route

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (1)

Filter routes

Both Edit routes

< 1 > ⚙

Destination	Target	Status	Propagated	Route Origin
120.0.0.0/16	local	Active	No	Create Route Table

Enter:

Destination: 0.0.0.0/0

Target: NAT Gateway

Assignment1-NAT

☰ VPC > Route tables > rtb-0dc7c8a6ce02c09a3 > Edit routes

Edit routes: rtb-0dc7c8a6ce02c09a3 / Private-RT

Destination	Target	Status
120.0.0.0/16	local	Active
0.0.0.0/0	local	-
	NAT Gateway	-
	nat-1f831c37c6e49c9e4	-

[Add route](#)

Clicked Save Changes.

Updated routes for rtb-0dc7c8a6ce02c09a3 / Private-RT successfully

rtb-0dc7c8a6ce02c09a3 / Private-RT

Details
Route table ID
rtb-0dc7c8a6ce02c09a3
VPC
vpc-0bc6fabf7f2bc9905 | Assignment1-VPC

Main
No
Owner ID
430776404757

Explicit subnet associations
-

Edge associations
-

[Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#) | [Related resources](#)

Routes (2)

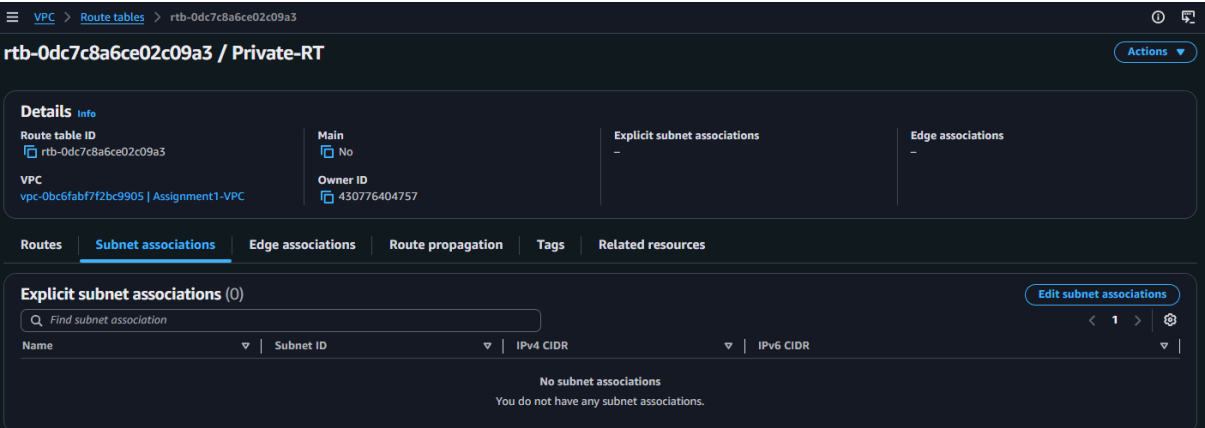
Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-1f831c37c6e49c9e4	Active	No	Create Route
120.0.0.0/16	local	Active	No	Create Route Table

route table:

Destination	Target
0.0.0.0/0	nat-1f831c37c6e49c9e4
120.0.0.0/16	local

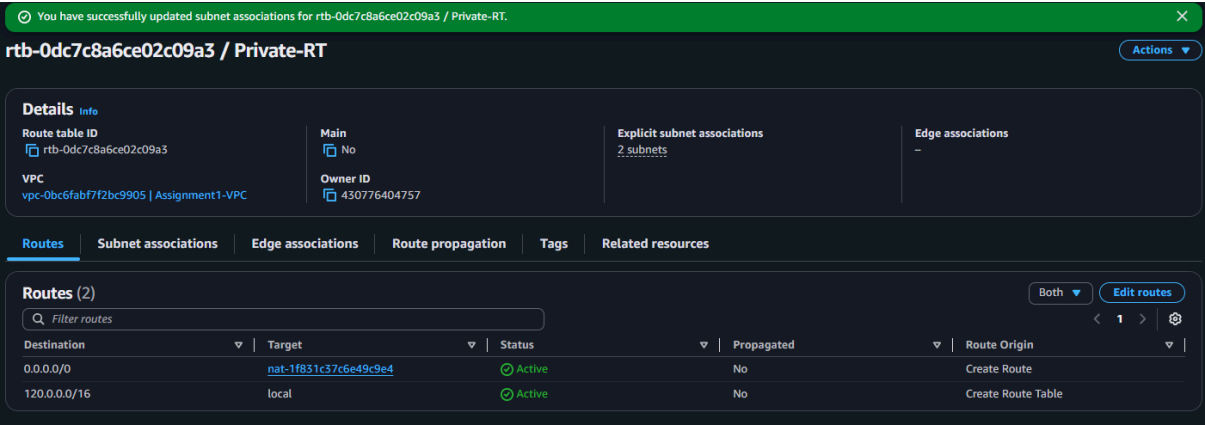
11. Associated private subnet with the route table

VPC → Route tables → Selected Private-RT → Subnet associations → Edit subnet associations



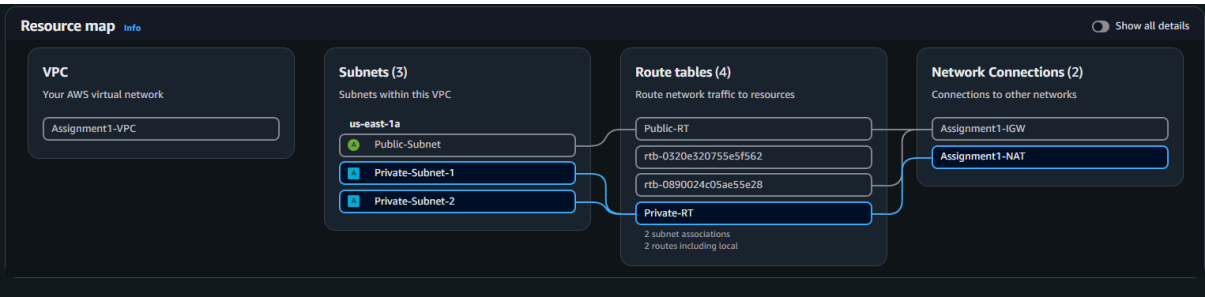
Selected: **Private-Subnet-1** and **Private-Subnet-2**

Clicked **Save associations**.



Both private subnets successfully use the NAT Gateway.

Verification:



The overall VPC resource map shows successful mapping of subnets, Gateway, Route tables.